THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 2: **Polynomials** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 28	$(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)$
2	**(a)** 30	$\alpha\beta(\alpha+\beta)$
3	**(a)** 7	$\sqrt{(\alpha+\beta)^2-4\alpha\beta}$
4	**(a)** $\frac{5}{2}$	$\frac{\alpha^2+\beta^2}{\alpha\beta}$
5	**(a)** 11	$\frac{c}{a}$ for the monic cubic
6	**(c)** 21	$(\alpha-\beta)^2=(\alpha+\beta)^2-4\alpha\beta$
7	**(d)** 7	Sum of zeroes $=9$, so other zero $=9-2$
8	**(d)** 72	$(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)$
9	**(a)** Both A and R are true and R is the correct explanation of A.	Each point where the graph meets the x-axis has $y=0$, i.e. $p(x)=0$; so R explains A.
10	**(c)** A is true but R is false.	Sum $=-\frac{b}{a}=\frac{8}{2}=4$, so A is true; R is false because $\frac{c}{a}$ is the product.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	New sum $=\alpha^2+\beta^2=10$ (1); product $=(\alpha\beta)^2=9$; polynomial $x^2-10x+9$ (1).	1+1
12	Sum $=6$, product $=k$ (1); $k=6$ (1).	1+1
13	Zeroes α and 3α: $4\alpha=8$ so $\alpha=2$ (1); $k=3\alpha^2=12$ (1).	1+1
14	New sum $=\alpha^2+\beta^2=13$ (1); product $=(\alpha\beta)^2=36$; polynomial $x^2-13x+36$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\alpha+\beta=3$, $\alpha\beta=2$ (1); sum $=(\alpha+\beta)+\frac{\alpha+\beta}{\alpha\beta}=\frac{9}{2}$ (1); product $=\alpha\beta+2+\frac{1}{\alpha\beta}=\frac{9}{2}$ (1).	1+1+1
16	$(\alpha+\beta)^2-(\alpha-\beta)^2=4\alpha\beta$ (1); $100-16=4c$ (1); $c=21$ (1).	1+1+1
17	$\sum\alpha=6$, $\sum\alpha\beta=11$ (1+1); $\sum\alpha^2=(\sum\alpha)^2-2\sum\alpha\beta=14$ (1).	1+1+1
	OR: $\alpha+\beta=5$, $\alpha\beta=6$ (1); sum $=(\alpha+\beta)+\frac{\alpha+\beta}{\alpha\beta}=\frac{35}{6}$ (1); product $=\alpha\beta+2+\frac{1}{\alpha\beta}=\frac{49}{6}$ (1).	1+1+1
18	$(\alpha+\beta)^2-(\alpha-\beta)^2=4\alpha\beta$ (1); $144-36=4c$ (1); $c=27$ (1).	1+1+1
	OR: $\sum\alpha=7$, $\sum\alpha\beta=14$ (1+1); $\sum\alpha^2=(\sum\alpha)^2-2\sum\alpha\beta=21$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $x^2-8x+15=(x-3)(x-5)$, so zeros are 3 and 5 (1). (ii) Sum $=8$, product $=15$ (1). (iii) Twice, at $(3,0)$ and $(5,0)$ (1). (iv) Reciprocal zeros $\frac{1}{3},\frac{1}{5}$: sum $=\frac{8}{15}$, product $=\frac{1}{15}$, polynomial $x^2-\frac{8}{15}x+\frac{1}{15}$, i.e. $15x^2-8x+1$ (2).	1+1+1+2
20	Let zeroes be $a-d, a, a+d$; sum $=3a=9$ so $a=3$ (2); product $=a(a^2-d^2)=15$ gives $d^2=4$ (2); zeroes $1,3,5$ (1).	2+2+1
	OR: $\alpha+\beta=2$, $\alpha\beta=-3$ (1+1); new sum $=-1$ (1); new product $=-6$ (1); polynomial x^2+x-6 (1).	1+1+1+1+1

Mistakes to avoid:

- Writing sum of zeros as $\frac{c}{a}$ and product as $-\frac{b}{a}$; it is sum $=-\frac{b}{a}$ and product $=\frac{c}{a}$.
- Forgetting the sign: for $x^2-8x+15$ the sum of zeros is $+8$, not -8.
- Counting the zeros of $y=p(x)$ as the y-intercept; zeros are where the graph meets the *x*-axis.
- Counting a parabola that only touches the x-axis as having no zeros or two different zeros; it has one repeated zero.
- Writing the polynomial from its zeros as $x^2+(\alpha+\beta)x+\alpha\beta$ instead of $x^2-(\alpha+\beta)x+\alpha\beta$.
- Claiming a quadratic polynomial always has two real zeros (e.g. x^2+1 has none).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME — SET 2

Chapter 2: **Polynomials** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** $\frac{37}{6}$	$\frac{\alpha^2+\beta^2}{\alpha\beta}$
2	**(d)** 26	$\frac{c}{a}$ for the monic cubic
3	**(a)** 35	$(\alpha-\beta)^2=(\alpha+\beta)^2-4\alpha\beta$
4	**(a)** 8	Sum of zeroes $=11$, so other zero $=11-3$
5	**(b)** 124	$(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)$
6	**(c)** 120	$\alpha\beta(\alpha+\beta)$
7	**(d)** 7	$\sqrt{(\alpha+\beta)^2-4\alpha\beta}$
8	**(b)** $\frac{5}{2}$	$\frac{\alpha^2+\beta^2}{\alpha\beta}$
9	**(d)** A is false but R is true.	$x^2+1\geq 1>0$, so its graph never meets the x-axis and it has no real zero: A is false; R is true.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both statements are true, but the degree is the highest power of x, which has nothing to do with R.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Sum $=9$, product $=k$ (1); $k=9$ (1).	1+1
12	Zeroes α and 4α: $5\alpha=10$ so $\alpha=2$ (1); $k=4\alpha^2=16$ (1).	1+1
13	New sum $=\alpha^2+\beta^2=26$ (1); product $=(\alpha\beta)^2=25$; polynomial $x^2-26x+25$ (1).	1+1
14	Sum $=11$, product $=k$ (1); $k=11$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\alpha+\beta=5$, $\alpha\beta=4$ (1); sum $=(\alpha+\beta)+\frac{\alpha+\beta}{\alpha\beta}=\frac{25}{4}$ (1); product $=\alpha\beta+2+\frac{1}{\alpha\beta}=\frac{25}{4}$ (1).	1+1+1
16	$(\alpha+\beta)^2-(\alpha-\beta)^2=4\alpha\beta$ (1); $196-64=4c$ (1); $c=33$ (1).	1+1+1
17	$\sum\alpha=9$, $\sum\alpha\beta=26$ (1+1); $\sum\alpha^2=(\sum\alpha)^2-2\sum\alpha\beta=29$ (1).	1+1+1
	OR: $\alpha+\beta=7$, $\alpha\beta=12$ (1); sum $=(\alpha+\beta)+\frac{\alpha+\beta}{\alpha\beta}=\frac{91}{12}$ (1); product $=\alpha\beta+2+\frac{1}{\alpha\beta}=\frac{169}{12}$ (1).	1+1+1
18	$(\alpha+\beta)^2-(\alpha-\beta)^2=4\alpha\beta$ (1); $256-100=4c$ (1); $c=39$ (1).	1+1+1
	OR: $\sum\alpha=7$, $\sum\alpha\beta=7$ (1+1); $\sum\alpha^2=(\sum\alpha)^2-2\sum\alpha\beta=35$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Sum of zeros $=30$ m (1). (ii) $x^2-30x+216=(x-12)(x-18)$, so breadth 12 m and length 18 m (1). (iii) Area $=$ product of zeros $=216\ \text{m}^2$ (1). (iv) Perimeter $=2(18+12)=60$ m (1); cost $=60\times 50=₹3000$ (1).	1+1+1+2
20	Let zeroes be $a-d, a, a+d$; sum $=3a=3$ so $a=1$ (2); product $=a(a^2-d^2)=-3$ gives $d^2=4$ (2); zeroes $-1, 1, 3$ (1).	2+2+1
	OR: $\alpha+\beta=2$, $\alpha\beta=-24$ (1+1); new sum $=-22$ (1); new product $=-48$ (1); polynomial $x^2+22x-48$ (1).	1+1+1+1+1

Mistakes to avoid:

- Writing sum of zeros as $\frac{c}{a}$ and product as $-\frac{b}{a}$; it is sum $=-\frac{b}{a}$ and product $=\frac{c}{a}$.
- Forgetting the sign: for $x^2-8x+15$ the sum of zeros is $+8$, not -8.
- Counting the zeros of $y=p(x)$ as the y-intercept; zeros are where the graph meets the *x*-axis.
- Counting a parabola that only touches the x-axis as having no zeros or two different zeros; it has one repeated zero.
- Writing the polynomial from its zeros as $x^2+(\alpha+\beta)x+\alpha\beta$ instead of $x^2-(\alpha+\beta)x+\alpha\beta$.

- Claiming a quadratic polynomial always has two real zeros (e.g. $x^2 + 1$ has none).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 2: **Polynomials** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 10	Sum of zeroes $= 13$, so other zero $= 13 - 3$
2	**(d)** 133	$(\alpha+\beta)^3 - 3\alpha\beta(\alpha+\beta)$
3	**(a)** -48	$\alpha\beta(\alpha+\beta)$
4	**(a)** 8	$\sqrt{(\alpha+\beta)^2 - 4\alpha\beta}$
5	**(b)** $\frac{29}{10}$	$\dfrac{\alpha^2+\beta^2}{\alpha\beta}$
6	**(b)** -8	$\frac{c}{a}$ for the monic cubic
7	**(a)** 15	$(\alpha-\beta)^2 = (\alpha+\beta)^2 - 4\alpha\beta$
8	**(a)** 8	Sum of zeroes $= 12$, so other zero $= 12 - 4$
9	**(a)** Both A and R are true and R is the correct explanation of A.	$S = 3 + (-2) = 1$, $P = -6$, so the polynomial is $x^2 - x - 6$; R is the rule used.
10	**(c)** A is true but R is false.	$x^2 - 4x + 4 = (x-2)^2$ has the single (repeated) zero 2, so A is true; R is false.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Zeroes α and 2α: $3\alpha = 12$ so $\alpha = 4$ (1); $k = 2\alpha^2 = 32$ (1).	1+1
12	New sum $= \alpha^2 + \beta^2 = 20$ (1); product $= (\alpha\beta)^2 = 64$; polynomial $x^2 - 20x + 64$ (1).	1+1
13	Sum $= 13$, product $= k$ (1); $k = 13$ (1).	1+1
14	Zeroes α and 4α: $5\alpha = 15$ so $\alpha = 3$ (1); $k = 4\alpha^2 = 36$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\alpha+\beta = 7$, $\alpha\beta = 10$ (1); sum $= (\alpha+\beta) + \dfrac{\alpha+\beta}{\alpha\beta} = \frac{77}{10}$ (1); product $= \alpha\beta + 2 + \dfrac{1}{\alpha\beta} = \frac{121}{10}$ (1).	1+1+1
16	$(\alpha+\beta)^2 - (\alpha-\beta)^2 = 4\alpha\beta$ (1); $400 - 36 = 4c$ (1); $c = 91$ (1).	1+1+1
17	$\sum\alpha = 3$, $\sum\alpha\beta = -6$ (1+1); $\sum\alpha^2 = (\sum\alpha)^2 - 2\sum\alpha\beta = 21$ (1).	1+1+1
	OR: $\alpha+\beta = 7$, $\alpha\beta = 6$ (1); sum $= (\alpha+\beta) + \dfrac{\alpha+\beta}{\alpha\beta} = \frac{49}{6}$ (1); product $= \alpha\beta + 2 + \dfrac{1}{\alpha\beta} = \frac{49}{6}$ (1).	1+1+1
18	$(\alpha+\beta)^2 - (\alpha-\beta)^2 = 4\alpha\beta$ (1); $324 - 16 = 4c$ (1); $c = 77$ (1).	1+1+1
	OR: $\sum\alpha = 9$, $\sum\alpha\beta = 24$ (1+1); $\sum\alpha^2 = (\sum\alpha)^2 - 2\sum\alpha\beta = 33$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) -1 and 3 (1). (ii) One zero, namely 2 (repeated), since the graph touches the x-axis at one point (1). (iii) $2x - 8 = 0 \Rightarrow x = 4$ (1). (iv) $p(x) = k(x+1)(x-3) = k(x^2 - 2x - 3)$; $p(0) = -3k = -6 \Rightarrow k = 2$ (1); $p(x) = 2x^2 - 4x - 6$ (1).	1+1+1+2
20	Let zeroes be $a-d, a, a+d$; sum $= 3a = 18$ so $a = 6$ (2); product $= a(a^2 - d^2) = 120$ gives $d^2 = 16$ (2); zeroes $2, 6, 10$ (1).	2+2+1
	OR: $\alpha+\beta = 3$, $\alpha\beta = -40$ (1+1); new sum $= -37$ (1); new product $= -120$ (1); polynomial $x^2 + 37x - 120$ (1).	1+1+1+1+1

Mistakes to avoid:

- Writing sum of zeros as $\frac{c}{a}$ and product as $-\frac{b}{a}$; it is sum $= -\frac{b}{a}$ and product $= \frac{c}{a}$.
- Forgetting the sign: for $x^2 - 8x + 15$ the sum of zeros is $+8$, not -8.
- Counting the zeros of $y = p(x)$ as the y-intercept; zeros are where the graph meets the *x*-axis.
- Counting a parabola that only touches the x-axis as having no zeros or two different zeros; it has one repeated zero.
- Writing the polynomial from its zeros as $x^2 + (\alpha+\beta)x + \alpha\beta$ instead of $x^2 - (\alpha+\beta)x + \alpha\beta$.

- Claiming a quadratic polynomial always has two real zeros (e.g. $x^2 + 1$ has none).

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